

⌈ Inequalities of Real Numbers.

⌈ Bernoulli's Inequality

If $a > -1$, $1 + na \leq (1+a)^n$ for all $n \in \mathbb{N}$.

Proof. If $n=1$, trivial. If the statement hold for $n=k \in \mathbb{N}$, then

$$(1+a)^{k+1} = (1+a)(1+a)^k \underset{1+a \geq 0}{\geq} (1+a)(1+ka) = 1 + (k+1)a + a^2 \geq 1 + (k+1)a.$$

⌈ Arithmetic-Geometric mean

If $a, b \in \mathbb{R}$, then $2ab \leq a^2 + b^2$ (Directly, it can be proved by $(a-b)^2 \geq 0$.)

⌈ Generalized A-G mean inequality,

If $a_i \geq 0$, then for all $n \in \mathbb{N}$,

$$(a_1 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} (a_1 + a_2 + \cdots + a_n)$$

proof. If $n=2$, $\sqrt{a_1 a_2} \leq \frac{1}{2}(a_1 + a_2)$, thus the statement holds.

Suppose that for $n=2^k$ the statement be hold. That is,

$$(a_1 \cdots a_n)^{\frac{1}{n}} \leq \frac{1}{n} (a_1 + \cdots + a_n)$$

$$\begin{aligned} \text{Now, } \frac{1}{2n} (a_1 + \cdots + a_{2n}) &= \frac{1}{2} \left(\frac{1}{n} (a_1 + \cdots + a_n) + \frac{1}{n} (a_{n+1} + \cdots + a_{2n}) \right) \\ &\geq \frac{1}{2} \left((a_1 \cdots a_n)^{\frac{1}{n}} + (a_{n+1} \cdots a_{2n})^{\frac{1}{n}} \right) \\ &= \frac{1}{2} \left(\left((a_1 \cdots a_n)^{\frac{1}{2n}} \right)^2 + \left((a_{n+1} \cdots a_{2n})^{\frac{1}{2n}} \right)^2 \right) \\ &\geq \frac{1}{2} \cdot 2 (a_1 \cdots a_n)^{\frac{1}{2n}} \cdot (a_{n+1} \cdots a_{2n})^{\frac{1}{2n}} \\ &= (a_1 \cdots a_n a_{n+1} \cdots a_{2n})^{\frac{1}{2n}} \end{aligned}$$

Therefore, by induction, the statement hold when $n=2^k$ ($k \in \mathbb{N}$).

To generalize, let $n \geq 2$ be given put $k \in \mathbb{N}$ s.t

$$2^{k-1} \leq n \leq 2^k \quad \frac{1}{2^k} \leq \frac{1}{n} \leq \frac{1}{2^{k-1}} \quad 1 \leq 2$$

put

$$b_i = \begin{cases} a_i & \text{if } i \leq n \\ \frac{1}{n} (a_1 + \cdots + a_n) = M & \text{if } n < i \leq 2^k \end{cases} \quad \frac{n}{2^k} \leq 1 \quad 0 \leq 1 - \frac{n}{2^k}$$

$$\text{Then, } (b_1 \cdots b_{2^k})^{\frac{1}{2^k}} \leq \frac{1}{2^k} \cdot (b_1 + \cdots + b_{2^k}) = \frac{1}{2^k} \cdot (nM + (2^k - n)M) = M$$

$$\text{And } (b_1 \cdots b_{2^k})^{\frac{1}{2^k}} = (a_1 \cdots a_n \cdot M^{2^k - n})^{\frac{1}{2^k}} = (a_1 \cdots a_n)^{\frac{1}{2^k}} \cdot M^{1 - \frac{n}{2^k}}$$

$$\text{Finally, } (a_1 \cdots a_n)^{\frac{1}{2^k}} \leq M^{\frac{n}{2^k}} \text{ implies } (a_1 \cdots a_n)^{\frac{1}{n}} \leq (a_1 + \cdots + a_n) \cdot \frac{1}{n}.$$

┌ Lagrange's Inequality.

Let $a_i, b_i \in \mathbb{R}$. Then,

$$\left[\sum_{i=1}^n a_i b_i \right]^2 = \left[\sum_{i=1}^n a_i^2 \right] \cdot \left[\sum_{i=1}^n b_i^2 \right] - \frac{1}{2} \cdot \sum_{1 \leq j, k \leq n} (a_j b_k - a_k b_j)^2$$